

YAN AN, China

WMO: 538450

Lat: 36.5757N

Lon: 109.4498E

Elev: 3875

StdP: 12.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	9.5	-13.2	3.1	19.1	-9.0	3.9	19.2	12.7	30.6	11.0	26.5	4.0	220	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	91.9	66.4	89.2	65.9	86.7	65.7	73.5	82.9	72.0	80.9	70.6	79.5	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.0	132.1	78.0	69.6	125.9	76.6	68.1	119.3	75.8	40.1	83.3	38.7	81.3	37.3	78.8	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.0	8.5	DB	-0.6	96.0	3.9	3.1	-3.4	98.2	-5.7	100.0	-7.9	101.7	-10.7	103.9	(4)
(5)			WB	-2.0	75.6	3.6	1.5	-4.6	76.6	-6.7	77.5	-8.8	78.4	-11.4	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	24.4	32.1	43.9	56.3	64.7	72.0	75.4	72.7	63.7	52.4	39.6	27.6	(6)
(7)		DBStd	18.59	6.50	7.68	8.05	7.67	6.05	4.83	4.46	4.67	5.91	6.05	6.78	6.34	(7)
(8)		HDD50	2600	792	502	222	27	1	0	0	0	0	46	316	694	(8)
(9)		HDD65	5606	1257	922	655	276	81	6	0	4	92	392	762	1159	(9)
(10)		CDD50	3385	0	0	32	216	455	659	788	702	410	119	4	0	(10)
(11)		CDD65	917	0	0	0	15	70	215	323	242	52	0	0	0	(11)
(12)		CDH74	8429	0	0	13	267	931	2163	2814	1793	427	21	0	0	(12)
(13)		CDH80	3029	0	0	1	68	264	856	1148	574	116	2	0	0	(13)
(14)	Wind	WSAvg	3.8	3.7	3.9	4.3	4.4	4.4	4.1	3.8	3.4	3.4	3.4	3.6	3.6	(14)
(15)	Precipitation	PrecAvg	20.6	0.2	0.3	0.7	1.1	1.7	2.4	4.8	4.1	3.0	1.5	0.7	0.1	(15)
(16)		PrecMax	33.9	0.6	1.0	2.1	3.3	4.6	6.7	19.1	7.6	6.9	3.2	2.4	0.7	(16)
(17)		PrecMin	14.4	0.0	0.0	0.0	0.2	0.4	0.8	1.0	1.4	0.7	0.2	0.0	0.0	(17)
(18)		PrecStd	4.3	0.2	0.2	0.5	0.7	1.0	1.3	3.1	1.8	1.5	0.9	0.7	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	60.2	75.3	87.8	90.0	95.2	95.8	92.1	89.4	76.6	65.8	50.2	(19)
(20)			MCWB	35.3	41.8	49.7	58.9	59.9	63.4	68.1	69.1	66.3	56.9	48.0	36.5	(20)
(21)		2%	DB	44.5	53.6	68.4	80.9	86.0	91.2	92.4	88.7	82.9	72.0	61.1	45.9	(21)
(22)			MCWB	32.6	38.2	47.1	55.0	58.5	63.5	68.1	69.0	64.5	54.6	45.5	34.0	(22)
(23)		5%	DB	40.3	49.8	64.0	76.4	82.9	88.1	89.5	85.9	78.6	68.3	56.3	42.4	(23)
(24)			MCWB	30.0	36.5	45.0	52.4	57.7	63.1	68.1	68.7	62.6	53.3	43.7	32.0	(24)
(25)		10%	DB	36.5	45.8	59.4	72.1	79.5	85.2	86.5	83.2	74.9	64.3	52.2	38.9	(25)
(26)			MCWB	27.8	34.6	42.7	50.6	56.6	62.5	68.4	68.1	61.5	51.5	42.0	30.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.2	42.9	52.1	61.2	64.8	70.9	75.6	75.0	73.1	60.3	51.2	37.6	(27)
(28)			MCD B	48.3	57.1	68.4	78.0	78.6	82.8	86.4	84.9	79.9	68.0	57.9	45.9	(28)
(29)		2%	WB	32.9	39.8	48.7	57.9	62.6	68.5	73.8	73.5	68.4	57.7	48.1	35.0	(29)
(30)			MCD B	42.9	50.8	65.5	73.4	77.3	80.8	83.9	82.1	77.0	66.0	58.2	44.1	(30)
(31)		5%	WB	30.5	37.7	46.2	55.4	61.0	66.9	72.4	72.0	65.4	55.8	45.5	32.9	(31)
(32)			MCD B	39.2	46.7	60.6	71.0	74.8	78.9	81.8	80.5	74.4	63.5	54.1	40.7	(32)
(33)		10%	WB	28.4	35.6	43.8	53.1	59.3	65.5	71.1	70.5	63.3	53.9	43.0	31.0	(33)
(34)			MCD B	35.5	44.6	57.4	66.8	73.1	77.6	80.0	78.8	71.4	61.9	51.2	37.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.0	21.3	22.3	23.4	23.5	22.2	18.6	17.7	18.0	20.4	20.2	19.8	(35)
(36)			MCDBR	27.4	28.1	30.0	30.9	30.6	28.9	25.0	23.1	25.0	27.5	28.1	25.4	(36)
(37)			MCWBR	18.1	16.2	14.5	12.6	11.4	8.9	7.1	7.4	9.2	12.5	15.2	16.0	(37)
(38)		5% WB	MCD BR	25.7	24.1	27.7	25.0	23.6	21.1	17.8	17.4	18.8	20.9	23.6	23.1	(38)
(39)			MCWBR	17.2	14.2	14.1	11.3	9.9	7.9	6.4	6.9	8.1	10.5	13.7	15.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.386	0.466	0.496	0.521	0.526	0.572	0.568	0.591	0.536	0.494	0.409	0.362	(40)	
(41)		taud	2.004	1.829	1.809	1.794	1.851	1.799	1.876	1.799	1.874	1.889	2.011	2.092	(41)	
(42)		Ebn at Noon	242	234	242	245	246	234	234	224	228	223	232	243	(42)	
(43)		Edh at Noon	42	57	63	68	65	69	63	67	59	53	41	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	889	1098	1484	1742	1904	1877	1723	1543	1245	1099	930	814	(44)	
(45)		RadStd	85	123	127	139	129	121	164	126	121	152	127	80	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	-0.65	N/A	N/A	N/A	-0.86	N/A	N/A	N/A	-234	-158	(46)	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-108	(47)	

Nomenclature: See separate page